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TECHNICAL SPECIFICATION

# Data Solutions (DATA);
# NGSI-LD MQTT Notification Binding

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# Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Data Solutions (DATA).

# Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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# Executive summary

The NGSI-LD API optionally supports delivering subscription notifications via MQTT instead of HTTP. Subscriptions are still created through the HTTP binding, but an MQTT endpoint URI can be specified to receive notifications. Notifications are sent as MQTT messages containing a JSON object with body (the actual notification) and metadata (Content-Type, @context link, and optional receiver-specific fields). The protocol supports MQTT v3.1.1 and v5.0, with configurable QoS levels (0–2) and MIME types. Default settings apply unless explicitly overridden via subscription parameters.

# Introduction

The present document defines the MQTT notification binding of the NGSI-LD API.

The NGSI-LD API and information model benefited from contributions developed in the context of the following research projects:

- No. 101136799 (MARCLAIMED)
- No. 101180452 (URBAQUANTUM)
- No. 101270878 (DYNACITY)
- No. 101081802 (CONSERWA)
- No. G69J18001180007 (Data Highway)
- No. FTE0000471 (GeminiPort)
- No. PE00000001 (DUSTAR, RESTART SPOKE 6)
- No. 101069732 (aerOS)
- No. 101135784 (ARISE)
- No. 101189589 (O-CEI)

and from contributions developed by members of the FIWARE® Community.

# 1 Scope

The present document defines the MQTT Notification binding for the NGSI-LD Core API specified in [1]. It supersedes prior versions in ETSI GS CIM 009 [i.1].

# 2 References

## 2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 104 175: "Data Solutions (DATA); NGSI-LD Core API".

[2] OASIS Standard: "MQTT Version 3.1.1 Plus Errata 01". Edited by Andrew Banks and Rahul Gupta. 10 December 2015.

[3] OASIS Standard: "MQTT Version 5.0". Edited by Andrew Banks, Ed Briggs, Ken Borgendale and Rahul Gupta. 07 March 2019.

[4] ETSI TS 104 176: "Data Solutions (DATA); NGSI-LD HTTP Binding".

[5] W3C® Recommendation 16 July 2020: "JSON-LD 1.1 - A JSON-based Serialization for Linked Data".

## 2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI GS CIM 009: "Context Information Management (CIM); NGSI-LD API".

[i.2] MQTT URI Scheme.

[i.3] IETF RFC 3987: "Internationalized Resource Identifiers (IRIs)".

[i.4] The Unicode Consortium: "Unicode® 15.0.0".

# 3 Definition of terms, symbols and abbreviations

## 3.1 Terms

For the purposes of the present document, the following terms apply:

NOTE 1: The letters "NGSI-LD" were added to most terms to confirm that they are distinct from other terms of similar/same name in use in other organizations, however, in the present document the letters "NGSI-LD" are generally omitted for brevity.

NOTE 2: The use of URI in the context of the present document also includes the use of International Resource Identifiers (IRIs) as defined in IETF RFC 3987 [i.3], which extends the use of characters to Unicode characters [i.4] beyond the ASCII character set, enabling the support of languages other than English.

## 3.2 Symbols

Void.

## 3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

| | |
|---|---|
| API | Application Programming Interface |
| ASCII | American Standard Code for Information Interchange |
| CIM | Context Information Management |
| HTTP | Hypertext Transfer Protocol |
| IETF | Internet Engineering Task Force |
| IRI | Internationalized Resource Identifier |
| JSON | JavaScript Object Notation |
| JSON-LD | JSON Linked Data |
| LD | Linked Data |
| MIME | Multi-purpose Internet Mail Extensions |
| MQTT | Message Queuing Telemetry Transport |
| NGSI | Next Generation Service Interfaces |
| QoS | Quality of Service |
| RFC | Request For Comments |
| TCP | Transport Control Protocol |
| TLS | Transport Layer Security |
| TS | Technical Specification |
| URI | Uniform Resource Identifier |

# 4 Introduction

The present document defines the MQTT notification binding of the NGSI-LD API, whose abstract operations are defined in ETSI TS 104 175 [1].

# 5 API MQTT Notification Binding

## 5.1 Introduction

This clause defines the optional support of the NGSI-LD API for sending notifications via the MQTT protocol [2] and [3]. The subscriptions are handled using the HTTP binding as specified in ETSI TS 104 176 [4], but instead of an HTTP endpoint, an MQTT endpoint is provided.

## 5.2 Notification behaviour

In case a subscription received via HTTP specifies an MQTT endpoint in the "notification.endpoint.uri" member of the subscription structure (defined by clauses 5.2.6.5.2, 5.2.6.7.3 and 5.2.6.6.1 of ETSI TS 104 175 [1]), and the MQTT notification binding is supported by the NGSI-LD implementation, notifications related to this subscription shall be sent via the MQTT protocol.

The syntax of an MQTT endpoint URI is `mqtt[s]://[][:]@[:]/[/]*` and follows an existing convention for representing an MQTT endpoint as a URI [i.2].

Username and password can be optionally specified as part of the endpoint URI. If the port is not explicitly specified, the default MQTT port is 1883 for MQTT over TCP and 8883 for MQTTS, i.e. Secure MQTT over TLS. MQTT supports the structuring of topics as a hierarchy with any number of subtopic levels, which can be specified as part of the endpoint URI.

In MQTT, all non-protocol information has to be included into the MQTT message. This means that the actual notification as specified in clause 5.2.6.9.1 of ETSI TS 104 175 [1], as well as additional information like MIME type, possibly the link to the *@context* and additional user-specified information, which in the HTTP case is provided as headers, has to be included into the MQTT message. The MQTT notification message shall be provided as a JSON Object with the two elements `"metadata"` and `"body"`. The actual notification, as specified in clause 5.2.6.9.1 of ETSI TS 104 175 [1] is the value of `"body"`, whereas any additional information is provided as key-value pairs in `"metadata"`.

For the MQTT protocol, there are currently two versions supported, MQTTv3.1.1 [2] and MQTTv5.0 [3]. Also, there are three levels of quality of service:

- at most once (0);
- at least once (1); and
- exactly once (2).

These can be specified in the subscription as part of the optional array of KeyValuePair type (defined by clause 5.2.6.6.4 of ETSI TS 104 175 [1]) "notification.endpoint.notifierInfo". The MQTT protocol parameters can be found in Table 5.2-1. If not present, the given default value is used.

**Table 5.2-1: Protocol parameters for MQTT in notifierInfo**

| Key | Possible Values | Default | Source | Description |
|---|---|---|---|---|
| MQTT-QoS | 0, 1, 2 | 0 | Subscription's *notification.endpoint .notifierInfo* | MQTT Quality of service, at most once (0), at least once (1) and exactly once (2) |
| MQTT-Version | mqtt3.1.1, mqtt5.0 | mqtt5.0 | Subscription's *notification.endpoint .notifierInfo* | Version of MQTT protocol |

The MIME type associated with the notification shall be `"application/json"` by default. However, this can be changed to`"application/ld+json"` by means of the "endpoint.accept" member. The MIME type is specified as Content-Type in the `"metadata"` element of the MQTT message. If the target MIME type is `"application/json"` then the reference to the JSON-LD *@context* is provided as Link in the `"metadata"` element of the MQTT message, following the specification of the HTTP Link header as mandated by the JSON-LD specification [5], section 6.2 (to the default JSON-LD *@context* if none available). Table 5.2-2 lists these "receiver side" metadata parameters.

**Table 5.2-2: Parameters for MQTT in "metadata"**

| Key | Possible Values | Default | Source | Description |
|---|---|---|---|---|
| Content-Type | `"application/json"`, `"application/ld+json"` | `"application/json"` | Subscription's *notification.endpoint.accept* | MIME type of the notification included in the `"body"` element of the MQTT message |
| Link | Same format as specified in JSON-LD specification [5], section 6.2 for the HTTP Link header | | Link Header provided in Subscription | Contains the reference to the *@context* in case Content-Type is `"application/json"`. Example: http://myhost.org/mycontext; rel="http://www.w3.org/ns/json-ld#context"; type="application/ld+json" |

Additionally, if the optional array of KeyValuePair type (defined by clause 5.2.6.6.4 of ETSI TS 104 175 [1]) *notification.endpoint.receiverInfo* of the subscription is present, then a new entry for each member named "key" of the key, value pairs that make up the array shall be generated and added to the `"metadata"` element of the MQTT message. The content of each entry shall be set equal to the content of the corresponding "value" member of the *KeyValuePair*.

# Annex A (informative): Change history

| Date | Version | Information about changes |
|---|---|---|
| January 2026 | 1.0.0 | Initialized with content taken from CIM009 (2025/07, V1.9.1) |
| September 2026 | 1.1.1 | First published version |

# History

| Version | Date | Status |
|---|---|---|
| V1.1.1 | September 2026 | Publication |
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